

Advanced Mathematical Techniques (AMT)

Notes

1 Today's stuff

Let's determine the value of :

$$\int_0^1 \frac{\ln(1-t)}{t} dt$$

Let's first figure out if it converges so we can see if we even need to compute it. The integrand diverges at 0 and 1. To compute at 0, we expand the logarithm:

$$\frac{-t - \frac{t^2}{2} - \frac{t^3}{3}}{t} \xrightarrow{ast \rightarrow 0byLHOP} -1$$

Now, at 1, there's a logarithmic divergence. However, if we remember from last class, logarithms are weaklings, so we can still integrate at 1. To fully compute the log, let's just do the exponential.

$$\int_0^1 \frac{1}{t} (1-t)^\varepsilon dt = \int_0^1 t^{-1} (1-t)^\varepsilon \xrightarrow{B_1} \frac{\Gamma(0)\Gamma(1+\varepsilon)}{\Gamma(1+\varepsilon)}$$

This appears as though everything is falling apart, with the cancellations of ε and the undefined Γ . However, let's keep looking.

$$\int_0^1 \frac{1}{t} (1-t)^\varepsilon dt = \int_0^1 t^{-1} e^{\varepsilon \ln(1-t)} dt = \sum_{k=0}^{\infty} \frac{\varepsilon^k}{k!} \int_0^1 t^{-1} \ln^k(1-t) dt = \int_0^1 t^{-1} dt + \varepsilon \int_0^1 t^{-1} \ln(1-t) dt + \dots$$

However, the first integral there destroys everything since it's undefined. We can, however, define a regulator to prevent the divergence from destroying stuff. To do so, let's change the power of t in the original integral to:

$$\int_0^1 \frac{1}{t^{1-\delta}} (1-t)^\varepsilon dt = \int_0^1 t^{-1+\delta} (1-t)^\varepsilon dt \xrightarrow{B_1} \frac{\Gamma(\delta)\Gamma(1+\varepsilon)}{\Gamma(1+\varepsilon+\delta)}$$

This prevents things from cancelling and helps us preserve information. Further,

$$Above = \frac{\delta\Gamma(\delta)\Gamma(1+\varepsilon)}{\delta\Gamma(1+\varepsilon+\delta)} = \frac{\Gamma(1+\delta)\Gamma(1+\varepsilon)}{\delta\Gamma(1+\varepsilon+\delta)}$$

Further, expanding:

$$\begin{aligned} \frac{1}{\delta} \cdot \frac{\Gamma(1+\delta)\Gamma(1+\varepsilon)}{\Gamma(1+\varepsilon+\delta)} &= \frac{1}{\delta} \exp\left(-\gamma\delta + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} \delta^k - \gamma\varepsilon + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} \varepsilon^k\right) - \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} (\varepsilon + \delta)^k \\ &= \frac{1}{\delta} \left[1 + \sum_{k=2}^{\infty} \frac{(-1)^k \zeta(k)}{k} + \dots \right] \end{aligned}$$

We can pull out the term with both ε, δ , which would give us the answer. However, we could also expand as such:

$$\int_0^1 t^{-1+\delta} (1-t)^\varepsilon dt = \int_0^1 t^{-1+\varepsilon} dt + \varepsilon \cdot \int_0^1 t^{-1+\varepsilon} \ln(1-t) dt + \frac{\varepsilon^2}{2!} \int_0^1 t^{-1+\delta} \ln^2(1-t) dt$$

Thus, by the above expansion, the value of the integral is $-\frac{\pi^2}{6}$

STEPS: 1. Check the bounds to make sure the integral converges 2. Sub in for epsilon/delta 3. Double check against the expansion for the coefficient that you need to multiply by the result of finding the right power in the expansion. 4. Solve and done.